

# RAG Engineering — Module 03

## Document Ingestion — Updated 11-Notebook Sequence

This is the updated sequence aligned with the current master RAG Engineering curriculum. Module 03 is an **11-notebook module** and sits between the baseline RAG Pipeline and Chunking.

### Module Goal

Teach how to turn real-world documents into clean, structured, validated content ready for chunking, embeddings, indexing and retrieval.

### 11-Notebook Sequence

| #  | Notebook                                          | Scope                                                                                                                                                                                           |
|----|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | Ingestion Overview                                | Ingestion architecture; document lifecycle; source registration; parser-first strategy; document types; provenance; where ingestion fits in RAG.                                                |
| 02 | Document Parsing                                  | General document parsing; real documents; extracting text and document elements; parser outputs; preserving basic structure.                                                                    |
| 03 | PDF Parsing                                       | PDF internals; PyMuPDF; text extraction; blocks, lines, spans and coordinates; images; tables; PyMuPDF4LLM; LLM-oriented Markdown; safe resource handling with <code>pymupdf.open(...)</code> . |
| 04 | OCR                                               | OCR fundamentals; image-to-text; standalone image OCR; scanned PDF workflow; rendering pages to images; confidence; OCR errors; OCR and tables; validation.                                     |
| 05 | Cleaning & Normalization                          | Whitespace and encoding cleanup; repeated headers/footers; normalization; conservative cleaning; preserving retrieval-relevant information.                                                     |
| 06 | Structure Preservation                            | Headings; paragraphs; lists; sections; reading order; preserving document hierarchy and semantic structure.                                                                                     |
| 07 | Metadata & Provenance                             | Document metadata; page/section provenance; source identifiers; timestamps/version information; traceability.                                                                                   |
| 08 | Tables & Structured Documents                     | Tables; structured records; layout-aware extraction; table semantics; difficult structured documents.                                                                                           |
| 09 | General Document Parsers                          | Unstructured and Docling; compare general-purpose parsing approaches; parser selection.                                                                                                         |
| 10 | Advanced Document Understanding                   | Advanced document understanding; VLM-assisted parsing; complex layouts; visually rich documents; escalation beyond conventional parsers/OCR.                                                    |
| 11 | Validation, Updates & Complete Ingestion Pipeline | Document validation; extraction quality checks; updates and versioning; changed documents; complete ingestion pipeline; output ready for chunking.                                              |

### Curriculum Progression

**Source documents → Parsing → PDF parsing → OCR → Cleaning → Structure → Metadata/Provenance → Tables → General parsers → Advanced document understanding → Validation/versioning → Complete ingestion pipeline → Chunking**

### Tool Boundaries

**PyMuPDF** provides low-level PDF inspection and extraction. **PyMuPDF4LLM** provides an LLM/RAG-oriented extraction path such as Markdown. **OCR** is image-to-text and is used when useful content exists as pixels. **Unstructured** and **Docling** provide broader document parsing, while VLM-assisted parsing is for cases where conventional parsing/OCR is insufficient.

## **Engineering Principle**

Use the cheapest method that successfully preserves the information needed downstream. Do not OCR every PDF and do not escalate to a VLM when conventional parsing is sufficient. Preserve content, structure, metadata and provenance, and make extraction failures visible.

## **Module Output**

The completed module produces a validated ingestion workflow that accepts real documents, selects an appropriate extraction path, preserves useful structure and provenance, handles updates/versioning, and produces content ready for Module 04 — Chunking.

## **Alignment with the Current Master Roadmap**

The current master roadmap explicitly requires Module 03 to use an 11-notebook structure and includes document parsing, PDF parsing, OCR, cleaning and normalization, structure preservation, metadata and provenance, tables and structured documents, Unstructured, Docling, advanced document understanding/VLM-assisted parsing, validation, document updates/versioning, and the complete ingestion pipeline.